

EC571: Digital VLSI Circuit Design

Homework 1 – Fall 2026

Total: 100 points

Due: September 23 at 11:59 pm ET

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 1 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all intermediate reasoning. A numerical answer, Boolean expression, or circuit diagram without explanation will receive limited credit.

For transistor-level questions, clearly indicate which transistors are ON/OFF for the input cases you analyze. For design questions, state your assumptions and justify your design choices.

You may use AI tools only for concept review. You may not use AI tools to produce final answers, final schematics, or final derivations. At the end of your submission, include one sentence disclosing whether and how you used AI tools.

Question 1: From Technology Scaling to Design Constraints [20 points]

This question asks you to reason about why VLSI design is not simply “make transistors smaller.”

A chip design team is comparing two technology generations. In the older generation, a digital block contains N transistors, has average switched capacitance C , operates at supply voltage V_{DD} , and runs at frequency f . In the newer generation, the team expects the following first-order changes:

$$N' = 2N$$

$$C'_{\text{per transistor}} = 0.65 C_{\text{per transistor}}$$

$$V'_{DD} = 0.8 V_{DD}$$

$$f' = 1.25f$$

Assume the activity factor is unchanged. Use the dynamic power approximation:

$$P_{\text{dyn}} = \alpha C_{\text{sw}} V_{DD}^2 f$$

where C_{sw} is the total switched capacitance.

(a) **Dynamic power scaling.** Derive the ratio

$$\frac{P'_{\text{dyn}}}{P_{\text{dyn}}}$$

Clearly show how transistor count, capacitance per transistor, voltage, and frequency each affect the result. [5 points]

Total switched capacitance is transistor count * capacitance / transistor

$$\therefore C_{\text{sw}} = N C_{\text{per transistor}}$$

$$\therefore \frac{P'_{\text{dyn}}}{P_{\text{dyn}}} = \frac{\alpha N' C'_{\text{per trans.}} (V'_{DD})^2 f'}{\alpha N C_{\text{per trans.}} V_{DD}^2 f} = \frac{\alpha 2N (0.65 C_{\text{per trans.}}) (0.8 V_{DD})^2 (1.25f)}{\alpha N C_{\text{per trans.}} V_{DD}^2 f}$$

$$= 2(0.65)(0.8)^2(1.25)$$

$$= 1.04$$

→ Newer gen. consumes 4% more power.

- (b) **Energy per operation.** Assume the number of useful operations per second scales proportional to frequency only. Derive the ratio of energy per operation in the newer generation relative to the older generation. [4 points]

$$\frac{E'_{op}}{E_{op}} = \frac{P'_{dyn} / P_{dyn}}{f' / f} = \frac{1.04}{1.25} = 0.832$$

- (c) **Power density.** Suppose the chip area scales by $0.7\times$, meaning

$$A' = 0.7A.$$

Using your result from part (a), derive the ratio of power density:

$$\frac{P'_{dyn}/A'}{P_{dyn}/A}.$$

[4 points]

$$\frac{P'_{dyn}/A'}{P_{dyn}/A} = \frac{P'_{dyn}}{A'} \cdot \frac{A}{P_{dyn}} = \frac{1.04}{0.7} = 1.486$$

- (d) **Design interpretation.** Based on your answers, explain whether the newer technology is automatically “better.” Your answer must discuss at least three of the following: performance, energy, power density, cooling, reliability, and design complexity. [4 points]

Not automatically better. Newer technology provides 25% higher performance, 16.8% lower dynamic energy/ops. yet it has 48.6% higher power density level. This causes heat removal to be much more demanding & potentially increase design complexity.

- (e) **Course connection.** Identify two topics from the first EC571 lectures that help address the design challenges revealed by this scaling exercise. For each topic, explain in one sentence why it matters. [3 points]

2 possible topics:

1) Dynamic Power & Voltage Scaling: Lowering voltage supply will reduce dynamic power, making it easier to manage the increased power density associated with scaling.

2) Transistor sizing: Adjust the transistor size to balance speed against total capacitance & energy consumption rather than maximizing performance at all cost.

Other relevant topics are also correct.

Question 2: Static CMOS Logic Implementation and Sizing [20 points]

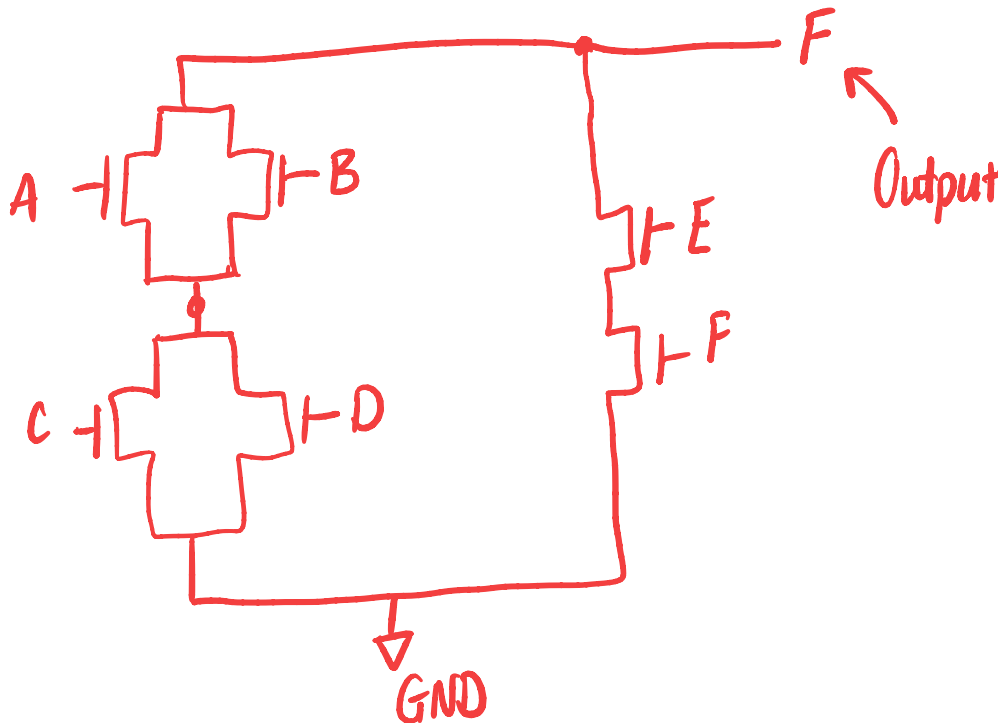
Implement the following Boolean function as a single static complementary CMOS gate:

$$F = \overline{(A + B)(C + D) + EF}.$$

Do not decompose the function into multiple NAND/NOR/inverter gates. You may assume true and complemented input signals are not both available unless you explicitly add an inverter. For this problem, implement the function directly using only inputs A, B, C, D, E, F .

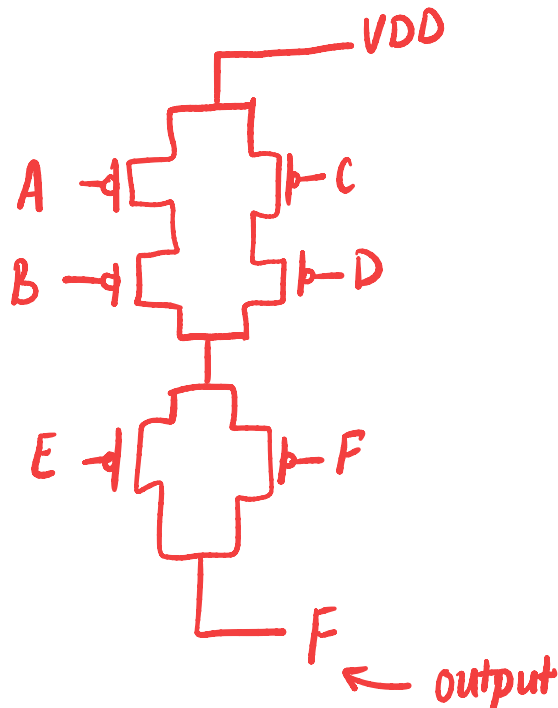
Assume:

- Unit width NMOS ON resistance is R_n .
 - Unit width PMOS ON resistance is $2R_n$.
 - Unit width NMOS input capacitance is C_g .
 - Unit width PMOS input capacitance is $2C_g$.
 - Transistor widths must be integer multiples of unit width.
 - Ignore diffusion capacitance in the numerical parts.
- (a) **Pull-down network.** Draw the NMOS pull-down network that causes $F = 0$. Clearly label all transistor gates and internal nodes. [4 points]



p "reverse"

- (b) **Pull-up network.** Draw the PMOS pull-up network as the dual of the pull-down network. Clearly label all transistor gates and internal nodes. [4 points]



- (c) **Correctness check using input vectors.** Choose one input vector for which $F = 0$. Identify the conducting path from V_{out} to ground. Choose one input vector for which $F = 1$. Identify the conducting path from V_{DD} to V_{out} . [4 points]

Example for $F = 0$: Input $(A, B, C, D, E, F) = (1, 0, 1, 0, 0, 0)$

$$F = \overline{(1+0)(1+0)} + 0 = 0$$

Conducting path : $F(V_{out}) \rightarrow n_A \rightarrow n_C \rightarrow GND$

Legend:

n_x : nMOS x
 p_x : pMOS x

Example for $F = 1$ Input $(A, B, C, D, E, F) = (0, 0, 1, 1, 0, 1)$

$$F = \overline{(0+0)(1+1)} + 0 = 1$$

Conducting path : $V_{DD} \rightarrow p_A \rightarrow p_B \rightarrow p_E \rightarrow F(V_{out})$

- (d) **Worst-case resistance before sizing.** Assuming all transistors are minimum size, identify the worst-case pull-down path resistance and the worst-case pull-up path resistance in terms of R_n . [4 points]

Pull down : Worst case pull down is path containing 2 series nMOS

$$R_{PD, \text{worst}} = 2R_n$$

Pull up : Worst case pull up is path containing 3 series pMOS.

$$R_{PU, \text{worst}} = (2R_n)3 = 6R_n$$

Each unit pMOS resistance : $2R_n$

- (e) **Transistor sizing.** Choose integer transistor widths so that the worst-case pull-down resistance is approximately R_n , and the worst-case pull-up resistance is approximately $2R_n$. Report the chosen width of every transistor.

Your answer must include a short explanation of how your sizing changes the input capacitance seen by each signal. [4 points]

Choose width of nMOS = 2 unit width & width of pMOS = 3 unit width

Pull-down : 2 series nMOS

$$R_{PD, \text{worst}} = \frac{R_n}{2} + \frac{R_n}{2} = R_n$$

$$W_{nA} = W_{nB} = W_{nC} = W_{nD} = W_{nE} = W_{nF} = 2$$

$$W_{pA} = W_{pB} = W_{pC} = W_{pD} = W_{pE} = W_{pF} = 3$$

Pull-up : 3 series pMOS

$$R_{PU, \text{worst}} = \frac{2R_n}{3} + \frac{2R_n}{3} + \frac{2R_n}{3} = 2R_n$$

Each input drives 1 nMOS & 1 pMOS so its capacitance increase

$$\text{from } 3C_g \text{ to : } C_{in} = 2C_g + 3(2C_g) = 8C_g$$

Question 3: VTC, Noise Margins, and Regeneration [20 points]

A CMOS inverter has the following measured DC transfer characteristic. Assume:

$$V_{DD} = 1.0V.$$

V_{in} (V)	V_{out} (V)
0.00	1.000
0.10	0.998
0.20	0.985
0.30	0.920
0.35	0.800
0.40	0.610
0.45	0.390
0.50	0.210
0.55	0.100
0.60	0.045
0.70	0.010
0.80	0.002
0.90	0.000
1.00	0.000

Use only the data in the table. Do not assume the inverter is symmetric.

- (a) **Switching threshold.** Estimate the switching threshold V_M , defined as the point where

$$V_{in} = V_{out}.$$

Use linear interpolation between the two closest table entries.

[4 points]

$$V_{in} = V_{out} \text{ between } (0.40, 0.610) \times (0.45, 0.390)$$

$$\therefore \text{slope} = m = \frac{0.390 - 0.610}{0.45 - 0.40} = -4.4$$

$$V_M = -4.4(V_M - 0.40) + 0.610$$

$$\therefore V_M = 0.439V$$

(b) **Local slope estimation.** Compute the approximate slope

$$\frac{\Delta V_{out}}{\Delta V_{in}}$$

for each interval between $V_{in} = 0.30V$ and $V_{in} = 0.60V$. Identify the interval with the largest magnitude slope. [4 points]

V_{in} interval	Slope
0.3 - 0.35	-2.4
0.35 - 0.40	-3.8
0.40 - 0.45	-4.4
0.45 - 0.50	-3.6
0.50 - 0.55	-2.2
0.55 - 0.60	-1.1

→ Largest interval slope : [0.40 - 0.45]

(c) **Noise-margin estimation.** Estimate V_{IL} and V_{IH} using the condition

$$\frac{dV_{out}}{dV_{in}} = -1.$$

Because the data is discrete, you must explain which intervals you use to estimate V_{IL} and V_{IH} . Then compute

$$NM_L = V_{IL} - V_{OL}$$

and

$$NM_H = V_{OH} - V_{IH}.$$

Use $V_{OH} = 1.0V$ and $V_{OL} = 0V$.

[5 points]

Estimate at each interval midpoint.

For V_{IL} : [0.2 - 0.3] : $m(0.25) = \frac{0.920 - 0.985}{0.3 - 0.2} = -0.65$
 [0.30 - 0.35] : $m(0.325) = \frac{0.800 - 0.920}{0.35 - 0.30} = -2.4$ } midpoint where slope first reaches -1
 $\therefore V_{IL} \approx 0.30V$

V_{IH} : [0.55, 0.6] , slope is -1.1
 [0.6, 0.7] , slope is -0.55 } $V_{IH} \approx 0.60V$

$$\therefore NM_L = V_{IL} - V_{OL} = 0.3 - 0 = 0.3V$$

$$NM_H = V_{OH} - V_{IH} = 1.0 - 0.6 = 0.4V$$

Any other reasonable estimation method is fine as well.

- (d) **Regeneration through a chain of inverters.** Suppose the input to the first inverter in a chain is

$$V_0 = 0.42V.$$

Use interpolation from the table to estimate

$$V_1, \quad V_2, \quad V_3,$$

where V_i is the output of the i -th inverter. Based on these values, explain whether the signal regenerates toward a legal logic value. [5 points]

Each inv. output becomes next inv. input

$$V_0 = 0.42V \rightarrow V_1 \rightarrow V_2 \rightarrow V_3$$

V_1 : Between $[0.40 - 0.45]$: $\frac{0.42 - 0.4}{0.45 - 0.4} = 0.4$ (40% of the way)

$$V_1 = 0.610 + 0.4(0.390 - 0.610) = 0.522V$$

V_2 : Same Idea $\rightarrow \frac{0.522 - 0.50}{0.55 - 0.50} = 0.44 \rightarrow V_2 = 0.210 + 0.44(0.1 - 0.210) = 0.1616V$

V_3 : $\frac{0.1616 - 0.10}{0.20 - 0.10} = 0.616 \rightarrow V_3 = 0.998 + 0.616(0.985 - 0.998) = 0.990V$

- (e) **Design interpretation.** If this inverter were used in a noisy environment, would you prefer a steeper or shallower VTC transition region? Explain your answer in terms of noise margins and regeneration. [2 points]

A steep transition means a small change in input would produce a large change in output, helping the next inverter to drive the signal towards either 0 or VDD.

For a similar switching threshold, also means narrowing the uncertain input region & improves noise margins.

Question 4: Inverter-Chain Delay Optimization with Non-Ideal Constraints [20 points]

A minimum-size inverter has input capacitance

$$C_{in,min} = 10fF.$$

You must drive a load

$$C_L = 6.4pF.$$

Use the normalized inverter delay model

$$d_i = f_i + \gamma$$

where

$$\gamma = 1.$$

For an inverter chain, the effective fanout of stage i is

$$f_i = \frac{C_{in,i+1}}{C_{in,i}}$$

for intermediate stages, and

$$f_N = \frac{C_L}{C_{in,N}}$$

for the final stage.

- (a) **Electrical effort.** Compute the total electrical effort

$$F = \frac{C_L}{C_{in,min}}.$$

[3 points]

$$6.4 \text{ pF} = 6400 \text{ fF}$$

$$\therefore F = \frac{6400}{10} = \boxed{640}$$

- (b) **Ideal number of stages.** Using the rule of thumb that a good stage effort is near 4, estimate the best number of inverter stages. Justify your answer. [4 points]

$$N \text{ stages} : 4^N \approx 640 \rightarrow N \approx \log_4 640 = 4.66$$

↳ Approx. 5 stages.

$$f = 640^{1/N}; \text{ bc } d_i = f_i + \gamma_i \text{ \& } \gamma = 1 \text{ so 1 inv. delay is } d_i = f_i + 1$$

$$\text{↳ sum of all delay} = D = \sum d_i = N(f+1)$$

therefore :

# of stages	fanout	Total delay
4	5.03	24.1
5	3.64	23.2
6	2.94	23.6

→ At $N=5$, optimal for reducing load each stage while not adding too much parasitic delay.

- (c) **Continuous sizing.** For your chosen number of stages N , compute the equal stage effort

$$f = F^{1/N}.$$

Then compute the input capacitance of each inverter stage in the chain, starting with the minimum-size inverter. [5 points]

$$f = F^{1/N} = 640^{1/5} = \boxed{3.6411}$$

stage	Input Cap.
1	10 fF → 10(3.6411)
2	36.411 fF → 36.411(3.6411)
3	132.58 fF x so on.
4	482.73 fF
5	1757.70 fF

→ Final inv w/extra load

$$f_5 = \frac{6400}{1757.70} \approx 3.6411$$

- (d) **Discrete sizing constraint.** Now assume that only the following inverter input capacitances are available:

$10fF, 20fF, 40fF, 80fF, 160fF, 320fF, 640fF, 1280fF, 2560fF.$

Choose a practical inverter chain using only these values. Compute the effective fanout of each stage and the total normalized delay. [4 points]

One possible 5 inv chain: $10 \rightarrow 40 \rightarrow 160 \rightarrow 640 \rightarrow 2560$

stage	Fanout f_i	Delay $d_i = f_i + 1$
1	$40/10 = 4$	5
2	$160/40 = 4$	5
3	$640/160 = 4$	5
4	$2050/640 = 4$	5
5	$6400/2560 = 2.5$	3.5

$$\text{Total delay} = 5 + 5 + 5 + 5 + 3.5 = 23.5$$

- (e) **Energy-delay design tradeoff.** Suppose a second design uses one extra inverter stage compared with your answer in part (d), and produces almost the same delay but uses larger total gate capacitance. Explain why a designer might reject the second design even if its delay is similar.

Your answer must discuss both dynamic energy and input loading on the previous logic stage. [4 points]

$$E_{\text{dynamic}} \propto C_{\text{sw}} V_{\text{DD}}^2$$

→ More switched capacitance → More energy to charge & discharge.

Therefore, 2nd design consumes more dynamic energy without any meaningful speed improvement.

For input loading, each gate input cap. is a load on the gate driving it → Enlarging gate → Increases that preceding stage's load.

Question 5: Logical Effort and Design Tradeoffs [20 points]

You need to design a logic path that computes a control signal Y and drives a large downstream load. The Boolean function is:

$$Y = \overline{ABCD}.$$

The signal A arrives from a minimum-size inverter whose input capacitance is:

$$C_{in,0} = 10 \text{ fF}.$$

The final output Y must drive:

$$C_L = 1.28 \text{ pF}.$$

Assume the following logical effort model:

$$d_i = g_i f_i + p_i$$

where g_i is the logical effort of gate i , f_i is the electrical fanout of gate i , and p_i is the parasitic delay of gate i .

Use the following logical-effort and parasitic-delay values:

$$\text{Inverter: } g = 1, p = 1.$$

$$n\text{-input NAND: } g = \frac{n+2}{3}, p = n.$$

$$n\text{-input NOR: } g = \frac{2n+1}{3}, p = n.$$

You are considering two implementations.

Implementation 1: Use one 4-input NAND gate to compute

$$Y = \overline{ABCD},$$

followed by a chain of inverters if needed.

Implementation 2: First compute two intermediate NAND outputs:

$$X_1 = \overline{AB}, \quad X_2 = \overline{CD},$$

then combine them using a 2-input NOR gate:

$$Y = \overline{X_1 + X_2}.$$

You may add inverters after Y as output buffers if needed. You may also insert an inverter on an internal signal only if you explain why it is logically required.

- (a) **Logic correctness.** Verify whether Implementation 1 and Implementation 2 compute the same function. For Implementation 2, explicitly simplify

$$\overline{AB + CD}$$

and state whether it equals $\overline{ABCD}$. If it does not, propose the smallest correction to Implementation 2 using NAND/NOR/inverter gates. [4 points]

$Y = \overline{ABCD} \rightarrow$ Implementation 1 is identical $Y = \overline{ABCD}$

Implementation 2: $Y = X_1 + X_2 = \overline{AB} + \overline{CD}$

$= \overline{ABCD}$ [De Morgan's]

$= (AB)(CD)$

$= ABCD \rightarrow$ not the same!

Solution: Add 1 inv after the final NOR gate

- (b) **Path logical effort.** For the corrected versions of the two implementations, identify the path from input A to output Y. Compute the path logical effort

$$G = \prod_i g_i$$

for each implementation before adding output-buffer inverters.

[4 points]

Implementation 1: $G_1 = g_{NAND4} = \frac{4+2}{3} = \boxed{2}$

Implementation 2: $NAND_2 \rightarrow NOR_2 \rightarrow INV$

$$G_2 = g_{imp.2} = \frac{4}{3} \left(\frac{5}{3} \right) (1) \approx \boxed{2.22}$$

Note: C, D is not on path from A so is not included

- (c) **Path effort and best number of stages.** For each implementation, compute the path electrical effort

$$F = \frac{C_L}{C_{in,0}}.$$

Assume no branching effort for Implementation 1. For Implementation 2, include any branching effort that appears on the path from A to Y . Then compute

$$H = GFB$$

and estimate the best number of stages using

$$N \approx \log_4 H.$$

Round N to a reasonable integer and justify your choice.

[4 points]

Electrical effort: $F = \frac{C_L}{C_{in,0}} = \frac{1280 \text{ fF}}{10 \text{ fF}} = 128$

Branch effort: $B = 1$

Implementation 1: $H_1 = G_1 F B_1 = 2(128)(1) = 256$

$N_1 \approx \log_4 H_1 = \log_4 256 = 4 \text{ stages}$

Implementation 2: $H_2 = G_2 F B_2 = \frac{20}{9}(128)(1) = 284.44$

$N_2 \approx \log_4 H_2 = \log_4 284.44 = 4 \text{ stages}$

- (d) **Gate sizing.** For the implementation you believe is better, choose gate input capacitances so that each stage has approximately equal stage effort

$$h = H^{1/N} \quad ; \quad D = \sum d_i \quad ; \quad d_i = g_i \frac{C_{out,i}}{C_{in,i}} + p_i$$

Work backward from the load and report the input capacitance of each stage on the path from A to Y. If you add output-buffer inverters, include them in the sizing. [5 points]

sizing backward: $h = g \frac{C_{out}}{C_{in}} \rightarrow C_{in} = \frac{g C_{out}}{h}$ / with $h = 4.10676$ for all stages

Final inv: $C_{inv} = \frac{1(1250)}{4.10676} = 311.68 \text{ fF}$

Drives 1250 fF, $g=1$

NOR2: $C_{NOR2} = \frac{5/3(311.68)}{4.10676} = 126.40 \text{ fF/input}$

NAND2: $C_{NAND2} = \frac{4/3(126.49)}{4.10676} = 41.07 \text{ fF/input}$

inv: $C_{inv} = \frac{1(41.07)}{4.10676} = 10.0 \text{ fF}$

Connected to A

Matches min-size driver, answer correctness confirmed

Note: The 4-stage count includes the existing minimum-sized inverter which drives A. Implementation 1 uses 2 output-buffer inverters which preserve the final output.

Implementation 2 uses 1 inverter after the NOR to obtain the required output logic.

Implementation 1: $h_1 = 256^{1/4} = 4$

$p_1 = 1 + 4 + 1 + 1 = 7$ (inv → NAND4 → inv → inv)

$D_1 = d_1 + d_2 + d_3 + d_4 = 1\left(\frac{40}{10}\right) + 1 + 2\left(\frac{10}{40}\right) + 4 + 1\left(\frac{320}{80}\right) + 1 + 1\left(\frac{1250}{320}\right) + 1 = 23$

Implementation 2: $h_2 = 284.444^{1/4} = 4.10676$

$p_2 = 1 + 2 + 2 + 1 = 6$ (inv → NAND2 → NOR2 → inv.)

$D_2 = d_1 + d_2 + d_3 + d_4 = 1\left(\frac{41.07}{10}\right) + 1 + \frac{4}{3}\left(\frac{126.4}{41.07}\right) + 2 + \frac{5}{3}\left(\frac{311.68}{126.4}\right) + 2 + 1\left(\frac{1250}{311.68}\right) + 1 \approx 22.43$

- (e) **Design-review recommendation.** Write a short design-review recommendation choosing between the two implementations. Your answer must discuss at least three of the following:

- logic correctness,
- number of stages,
- logical effort,
- parasitic delay,
- branching effort,
- input capacitance seen by signal A,
- robustness of using a large complex gate versus smaller gates.

A purely qualitative answer such as "fewer stages is better" will receive limited credit. [3 points]

Choose fixed implementation 2. Both versions use 4 stages with no branching penalty. Even though implementation 2 has higher logical effort, it has smaller parasitic delay (22.427 vs. 23). Final inverter is needed for correctness, where A has a larger input capacitance of 41.07 fF vs. 40 fF for implementation 1.

